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Reduction of a thin chromium oxide film on Inconel surface upon treatment with hydrogen plasma

Alenka Vesel [a,*], Miran Mozetic [a], Marianne Balat-Pichelin [b]

[a] Jozef Stefan Institute, Jamova cesta 39, 1000 Ljubljana, Slovenia
[b] PROMES-CNRS Laboratory, 7 Rue du four solaire, 66120 Font Romeu Odeillo, France

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ABSTRACT

Inconel samples with a surface oxide film composed of solely chromium oxide with a thickness of approximately 700 nm were exposed to low-pressure hydrogen plasma at elevated temperatures to determine the suitable parameters for reduction of the oxide film. The hydrogen pressure during treatment was set to 60 Pa. Plasma was created by a surfaguide microwave discharge in a quartz glass tube to allow for a high dissociation fraction of hydrogen molecules. Auger electron depth profiling (AES) was used to determine the decay of the oxygen in the surface film and X-ray diffraction (XRD) to measure structural modifications. During hydrogen plasma treatment, the oxidized Inconel samples were heated to elevated temperatures. The reduction of the oxide film started at temperatures of approximately 1300 K (considering the emissivity of 0.85) and the oxide was reduced in about 10 s of treatment as revealed by AES. The XRD showed sharper substrate peaks after the reduction. Samples treated in hydrogen atmosphere under the same conditions have not been reduced up to approximately 1500 K indicating usefulness of plasma treatment.

1. Introduction

Oxidation resistant alloys such as stainless steels, Inconels and alike usually contain a considerable amount of chromium. Upon heating in oxygen-containing atmosphere a thin oxide film grows on the surface [1,2]. The oxide film often contains compact chromium oxide. The Cr_2O_3 film represents a passive film on the alloy surface and prevents further growth of the oxide film [3,4]. Such a chromium oxide film is therefore often beneficial; however, in particular cases, it is important to remove it in order to keep the original surface properties such as electrical and thermal conductivity as well as visual appearance. The removal of the oxide film is often performed by chemical etching or mechanical treatments such as brushing and polishing. The chemical etching is ecologically unsuitable, whereas the mechanical removal is often not feasible because it is difficult to remove the oxide film from the surface of three-dimensional objects, especially those of a complex shape. Heating in hydrogen atmosphere definitely causes reduction of the oxide film; however, the required temperatures are too high for practical application [5–7]. Furthermore, it is often difficult to assure a high purity of hydrogen in the high-temperature furnace. Usually, small concentrations of oxidizing gases such as water vapour persist, and according to the Ellingham diagram the impurities prevent reduction at moderately high temperatures.

An alternative to heating in hydrogen atmosphere is application of hydrogen plasma [5,7,8]. Hydrogen molecules dissociate and partially ionize under plasma conditions, which modifies the reduction kinetics and lowers the temperature of the samples needed for effective reduction of the oxide film [7]. Application of such treatment has been demonstrated by several authors [5,7–17]. They studied reduction of Cu_2O, CuO, Fe_2O_3, Cr_2O_3, WO_3, MoO_3, Ta_2O_5, Nb_2O_5, Al_2O_3, TiO_2 and ZrO_2. A good review was given by Sabat et al. [7]. Most of the oxides were reduced to metal, except for Al_2O_3, TiO_2 and ZrO_2 they found no reduction or just reduction to their sub-oxides [7]. Reduction rate depended on metal-oxide binding energy. It is known that oxides which lay in the lower part of Ellingham diagram are too stable to allow significant reduction [7]. Hydrogen plasma was also applied for treatment of old Indian coins from nickel, which were oxidized and contaminated with organic and inorganic material [18].

From literature we can conclude that reduction was investigated only for pure metal-oxides, while there is a lack of information for reduction of oxides from alloys. Recently, we have already reported successful reduction of oxides from the stainless steel surface [8]. Although the oxidized stainless steel surface was composed mostly from chromium oxide, the surface after reduction was not rich with

* Corresponding author.
E-mail address: alenka.vesel@guest.arnes.si (A. Vesel).

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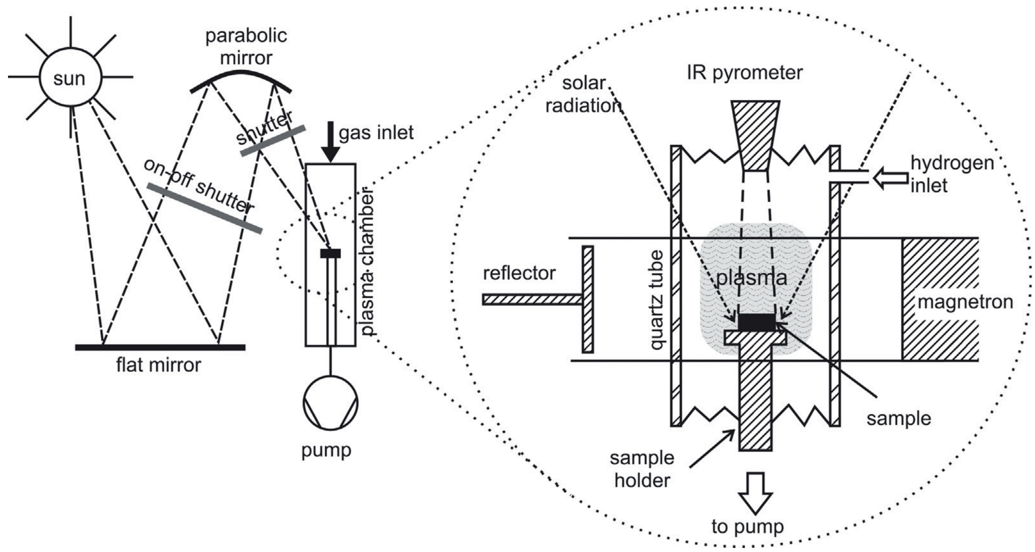

Fig. 1. Experimental set-up MESOX at the focus of the 6 kW Odeillo solar furnace.

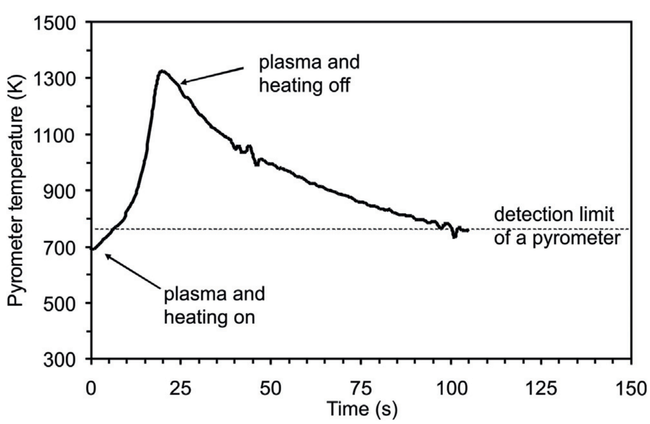

Fig. 2. Temperature profile of a sample during oxidation.

chromium but it was the same as the bulk material with the same stoichiometric composition. In the present paper we present results for reduction of oxides from Inconel alloy.

2. Experimental details

2.1. Sample treatment

Samples of Inconel 625 alloy (main components: Ni 58%, Cr 20–23%, Mo 8–10%, Fe 5%, Nb 3–4%) were polished and oxidized to obtain a thin film of chromium oxide. The oxidation was performed in oxygen plasma at elevated temperatures using the MESOX set-up (Fig. 1) [19]. After turning on oxygen plasma, the sample was additionally heated to 1300 K, which took about 20 s. When this temperature was reached, the plasma and heating were simultaneously turned off. A typical temperature profile of the sample during oxidation is shown in Fig. 2. After oxidation, the samples were exposed to hydrogen plasma in the same experimental system, as in Fig. 1 to study the reduction mechanisms [19]. The plasma reactor was a quartz glass tube of an inner diameter 5 cm and a length 50 cm. The reactor was pumped with a two-stage rotary pump of a nominal pumping speed 35 $m^3\,h^{-1}$ and ultimate pressure below 1 Pa. The pressure in the plasma reactor was measured with an absolute vacuum gauge. Commercially available hydrogen was leaked into the plasma reactor through a flowmeter calibrated for hydrogen. The experiments were performed at the pressure of 40 Pa corresponding to the gas flow of 8.5 $L\,h^{-1}$ (≈142 sc cm). All connections between vacuum components were with standard KF flanges using rubber gaskets. The vacuum system was frequently opened and never baked. Plasma was created in the middle of the reactor using a microwave discharge. The quartz glass tube was placed perpendicularly to the waveguide. The waveguide was connected to the magnetron source of microwaves (MW). The MW generator operated at variable powers up to 1200 W. The experiments were performed at the nominal power of 500 W, whereas the real power dissipated into plasma was estimated to approximately 350 W. The discharge power was too low to heat samples to high temperatures. In order to perform treatment under such conditions the samples were simultaneously exposed to concentrated solar radiation. The concentrated solar flux was focused onto a spot of a diameter 1 cm and taking into account an incident solar flux of 950 $W\,m^{-2}$, the concentrated solar flux received by the sample, for a shutter opening around 35°, was estimated to be approximately 150 W. The sample temperature was measured continuously upon heating by concentrated solar radiation using an infrared pyrometer. The pyrometer operated at the wavelength of 5 μm and was focused onto the sample's centre so that the radiation was collected from a spot of a diameter of 6 mm. The pyrometer signal was measured in mV. The corresponding temperature was calculated taking into account the normal spectral emissivity (ε) of 0.85 for the oxidized and 0.25–0.30 for fully reduced sample [20,21]. For partially reduced samples, the emissivity was somewhere in between; therefore, the emissivity of 0.60 was taken for approximation.

2.2. Surface characterization

Samples were characterized by Auger electron depth profiling (AES). AES depth profiling was performed using the spectrometer PHI SAM, model 545A. Samples were excited with a 3 keV electron beam of 1.0 μA current and 40 μm spot size. In order to obtain depth profiles they were sputtered with two symmetrically inclined Ar ion guns. Ions of kinetic energy 3 keV were rastering over a spot area of 3.5 mm × 3.5 mm. The etching rate was measured on standard Cr/Ni multilayer structures and was 10 nm/min. The pressure in the analysing chamber was below 2×10^{-9} mbar.

Any modification of the sample crystallinity was determined by X-ray diffraction. XRD analysis was performed using a PANalytical X'Pert Pro diffractometer (MPD) operating at 40 kV and 20 mA

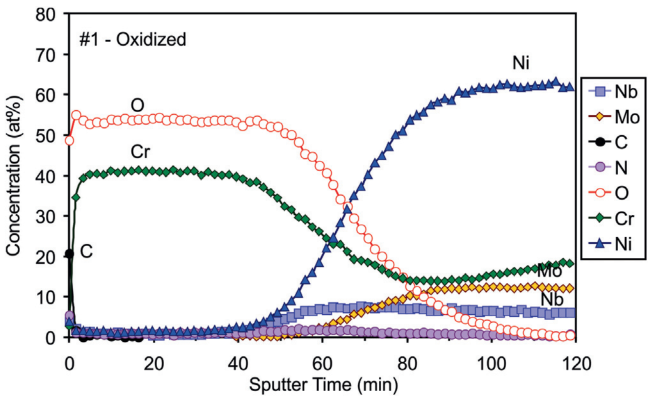

Fig. 3. AES depth profile of the oxidized sample #1.

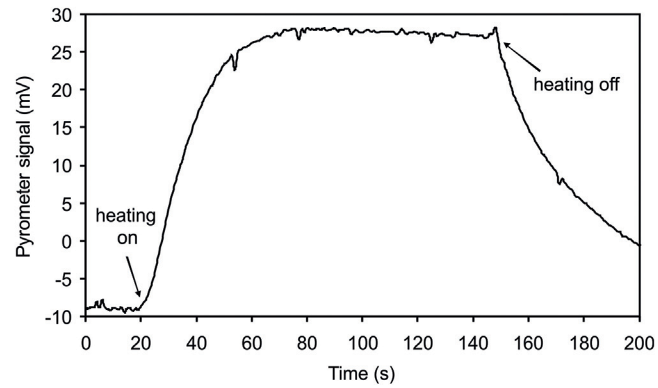

Fig. 4. Pyrometer signal upon heating of the oxidized sample in hydrogen without plasma.

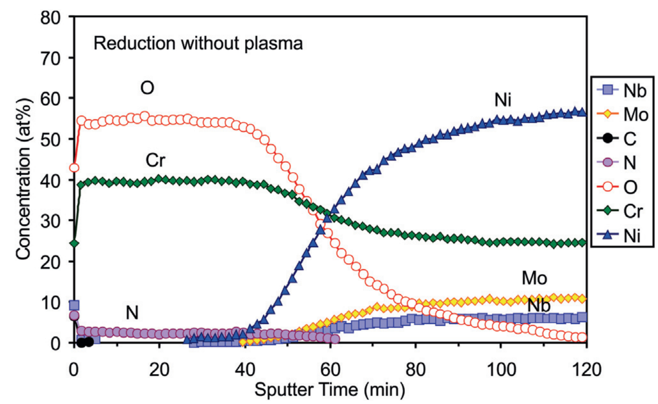

Fig. 5. AES depth profile of the sample treated in hydrogen atmosphere without plasma.

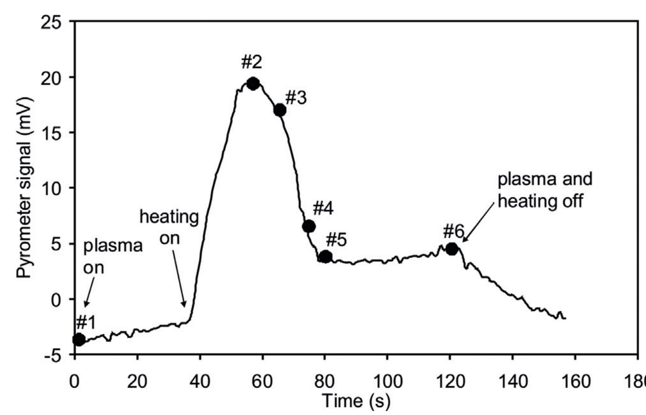

Fig. 6. Pyrometer signal upon heating of the oxidized sample in hydrogen with plasma.

(Cu Kα radiation, $\lambda = 0.15418$ nm). The X-ray diffraction measurements of θ–θ symmetrical scans were made in the range 10–100°. The step size and the time per step were respectively fixed at 0.01° and 5 s. The X-ray diffraction spectra were recorded with a PANalytical HighScore Plus software. Crystalline phases were identified by comparison with standard reference patterns (Powder diffraction file PDF-2, International Centre for Diffraction Data (ICDD)).

Evolution of the surface morphology of the samples during oxide reduction was investigated by scanning electron microscopy. JEOL JSM-7600F microscope was used.

The surface of the samples was also analysed with XPS instrument TFA XPS Physical Electronics. The base pressure in the XPS analysis chamber was about 6×10^{-10} mbar. The sample was excited by X-rays over a 400 μm^2 spot area with a monochromatic Al $K_{\alpha 1,2}$ radiation at 1486.6 eV. Photoelectrons were detected with a hemispherical analyzer positioned at an angle of 45° with respect to the normal to the sample surface. Survey scan spectra were made at a pass energy of 187.85 eV, while the individual high-resolution spectra were taken at a pass energy of 58 eV and 0.125 eV energy step. Sputtering for depth profiling of the oxidized layer was performed with Ar^+ ion beam with the energy of 3 keV at an incidence angle of 45° and a raster of 3 mm × 3 mm. This resulted in the sputtering rate of maximum 2 nm/min. XPS concentration profiles were evaluated using relative sensitivity factors from the manufacturer.

3. Results and discussion

An AES depth profile of an as-oxidized sample is presented in Fig. 3. The elemental composition is presented versus the argon-ion sputtering time. As mentioned in the experimental section the sputtering rate was approximately 10 nm/min. There is an oxide film of thickness approximately 700 nm on the sample surface. The depth profile reveals nearly pure chromium oxide. There is a rather broad interface between the oxide film and the bulk metal. The interface can be explained by the increased roughness due to oxidation process rather than an interface rich in several compounds.

A sample was treated in hydrogen atmosphere at the pressure of 40 Pa only by solar radiation (without turning on plasma). The pyrometer signal is plotted in Fig. 4. The temperature increased monotonously with treatment time. For the first several seconds the increase was nearly linear with the slope of 1.2 mV/s. Taking into account the emissivity of the oxidized sample of $\varepsilon = 0.85$ at the wavelength of 5 μm the temperature rise was approximately 30 K/s [2]. At elevated temperature the slope of the curve presented in Fig. 4 was less steep, and after about a minute it becomes rather constant at approximately 1500 K ($\varepsilon = 0.85$). The monotonous curve indicates no reduction of the oxide film at the experiment with hydrogen at equilibrium conditions, i.e. without plasma. The AES depth profile for this sample (Fig. 5) confirmed this conclusion because it was very similar to the depth profile of untreated sample, which is presented in Fig. 3. This result therefore clearly shows that neutral hydrogen atmosphere at elevated temperatures cannot cause the oxide reduction.

All other experiments were performed in hydrogen plasma. Fig. 6 represents the behaviour of the pyrometer signal of a sample treated in hydrogen plasma for 2 min. The curve is now completely different from the one corresponding to heating without plasma. The pyrometer signal first increases rather linearly with the slope (1.3 mV/s) similar to the one presented in Fig. 4; however, after 40 s a rather sharp peak is observed. After the maximum, the pyrometer signal dropped to approximately 4 mV and remained nearly

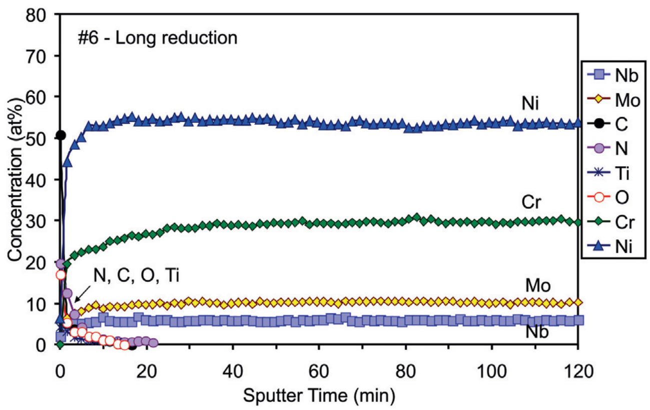

Fig. 7. AES depth profile of the sample #6 heated in hydrogen plasma for 2 min.

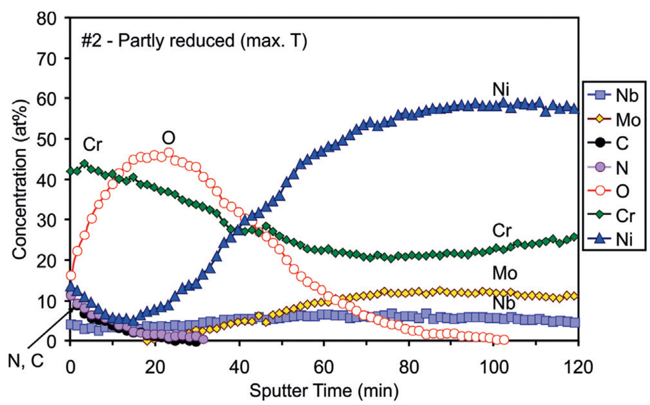

Fig. 8. AES depth profile of the sample #2 heated in hydrogen plasma until the inflection point on the pyrometer signal was detected.

unchanged until the heating by concentrated solar radiation was turned off. This point is marked #6 in Fig. 6. As mentioned earlier the solar flux was constant. The appearance of the peak, therefore, cannot be explained by a sudden decrease of the heating power but rather by the change in the sample emissivity. Namely, the emissivity of pure (un-oxidized) alloy is much lower than for the oxidized alloy [20,21]. The maximum signal/temperature reached before the sudden decrease is ~20 mV corresponding to approximately 1300 K for a normal spectral emissivity of 0.85, and on the temperature plateau of 4 mV after reduction of the oxide layer, the temperature is in between 1210 and 1270 K taking into account a normal spectral emissivity of 0.30–0.25 that is correct for the pure alloy. The inflection point immediately after the peak should reveal the reduction of the surface oxide film. Fig. 7 confirms this explanation because there is practically no oxygen in the AES depth profile.

The reduction kinetics was determined by measuring depth profiles of samples which were treated for different times. The samples are marked with numbers in Fig. 6. Sample #1 corresponds to the untreated one. Sample #2 was heated only until the peak is observed, sample #3 just after the peak, sample #4 before the constant pyrometer signal was observed and sample #5 just after it. The AES depth profiles of these samples are shown in Figs. 7–11.

As already mentioned, the sample #2 was heated in hydrogen plasma until the inflection point in the pyrometer signal became apparent. Unlike untreated sample #1 whose depth profile is shown in Fig. 3, the sample #2 has reduced concentration of oxygen close to the surface, which is revealed from Fig. 8. The chromium concentration right on the surface is still about 40 atomic% which is typical for chromium oxide, however, there is an increased concentration of nickel. The initial step in reduction of the oxide film is there-

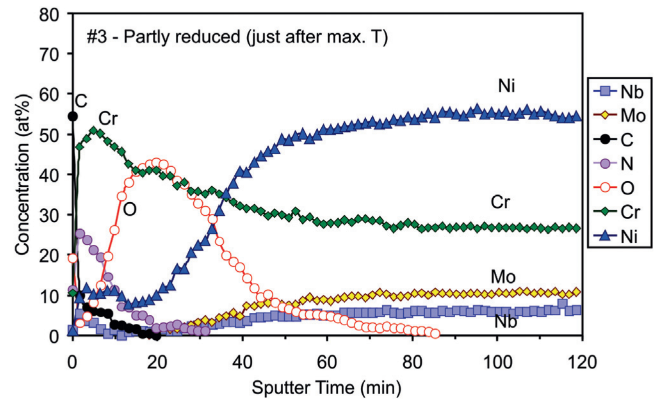

Fig. 9. AES depth profile of the sample #3 where heating was turned off just after reaching the maximum temperature.

fore also associated to migration of nickel through the oxide film. The thickness of the oxide film is somehow reduced indicating also migration of oxygen from the interface between the oxide and the bulk alloy towards the surface.

The reduction of the oxide film progresses rapidly after the high-enough temperature is obtained. Fig. 9 represents the depth profile of the sample #3 which was treated until the first inflection point on the pyrometer curve after the peak. Although the treatment time was only about 8 s longer than for sample #2, the concentration of oxygen is much lower. The concentration profile for oxygen indicates extensive reduction in the surface film of thickness approximately 100 nm and also migration of oxygen from the interface towards the surface. The AES depth profiles cannot reveal the lateral uniformity of oxygen concentration, because the lateral resolution of the Auger electron spectrometer is rather poor. Thus no attempt was made to determine any lateral distribution of oxygen or other elements. The exact migration mechanisms therefore cannot be stated; however, it is apparent that oxygen migrated towards the surface where extensive reduction occurred. The reduction cannot be explained by interaction with hydrogen molecules because reduction does not occur in hydrogen atmosphere at thermal equilibrium. The reactive hydrogen particles involved are either positively charged ions or neutral atoms in the ground state. All other particles can be neglected because there are practically no metastables in hydrogen plasma, because both highly excited molecules and atoms lose their potential energy by radiation. The charged particles are definitely more reactive than neutral hydrogen atoms because they also have kinetic energy gained by passing the sheath between bulk plasma and the surface. The density of ions in plasma created by the discharge used for our experiments is of the order of 10^{17} m^{-3}, thus the corresponding flux of ions onto the surface in the plan-parallel geometry is a few times 10^{19} m^{-2} s^{-1}. If all hydrogen ions capture oxygen from the sample surface, the removal rate would be a few monolayers per second, thus approximately 1 nm/s. The observed rate is much larger. Taking into account AES depth profiles presented in Figs. 8 and 9 the rate is an order of magnitude larger. One possible explanation involves neutral hydrogen atoms whose density is several orders of magnitude larger in our plasma [21]. Another possible explanation for the rapid removal of chromium oxide from the surface could be the formation of hydroxides of a rather high vapour pressure, which immediately desorb from the surface at elevated temperature [22]. Such mechanisms were already explained in our previous paper, where we investigated the oxide reduction from the stainless steel surface [23]. At high temperature, hydrogen can interact with chromium oxide and form $Cr(OH)_3$, which is desorbed from the surface after formation [23]. Another possible

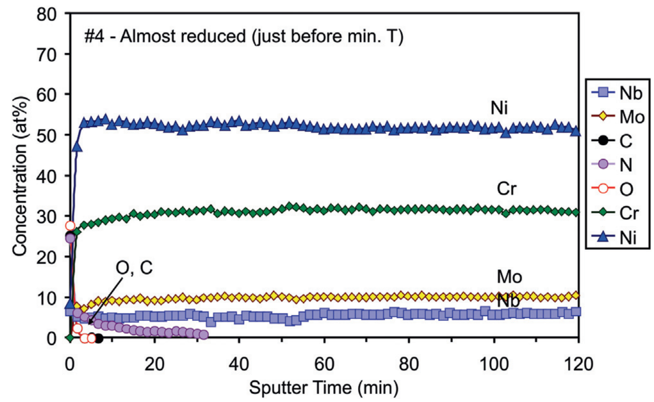

Fig. 10. AES depth profile of the sample #4 where heating was turned off just before reaching the minimum temperature.

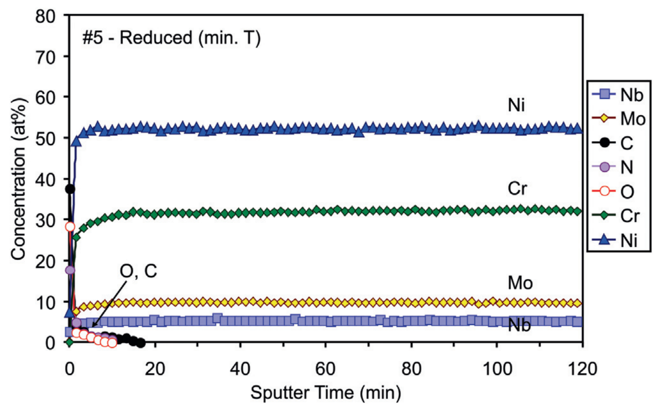

Fig. 11. AES depth profile of the sample #5 where heating was turned off just when the minimum temperature was reached.

compound could be also formation of $CrO_2(OH)_2$ when chromium oxide interacts with the water vapour [23].

Anyway, the reduction occurs fast, once it starts and leads to complete reduction of the oxide film. Fig. 10 reveals AES depth profile of the sample #4. No oxygen is observed in this depth profile apart from the very surface which is likely due to formation of the native oxide film upon exposure to air. Here it is worth mentioning that the samples were kept in hydrogen atmosphere for several minutes after turning off the heating to prevent post-treatment oxidation. The depth profiles of samples #4 and #5 (Figs. 10 and 11) are very similar and resemble to those of the bulk Inconel.

AES results are further supported with XPS results, which are shown in Fig. 12. This figure shows high resolution peaks of the

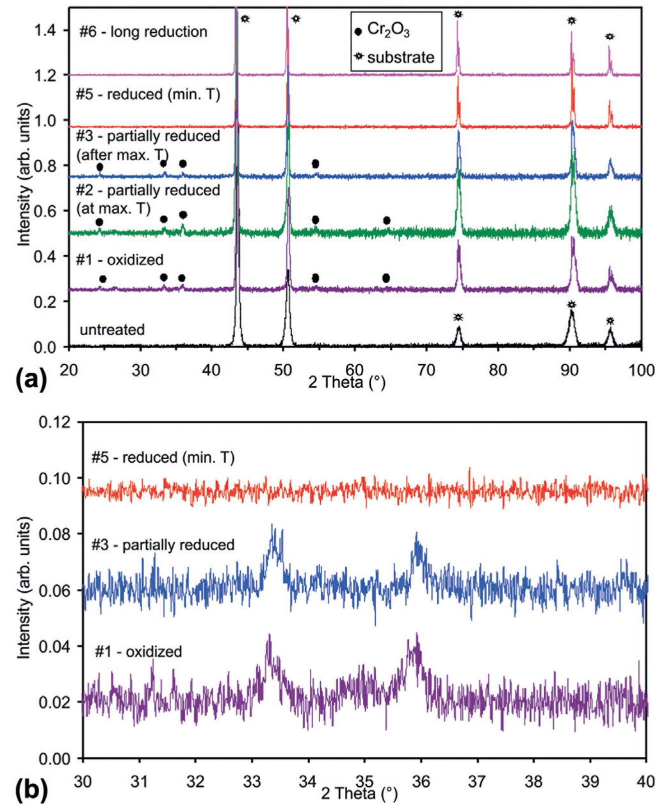

Fig. 13. (a) Normalized XRD patterns of samples and (b) zoom for 3 patterns showing the reduction process.

most important elements from Inconel i.e. Cr 2p, Ni 2p and Nb 3d for the oxidized sample (#1), partly reduced sample (#3) and reduced sample (#5). The spectra were recorded after few minutes of etching to remove the surface contamination. Fig. 12a shows chromium Cr $2p_{3/2}$ and Cr $2p_{1/2}$ peaks for the oxidized sample (#1), which are located at 576.6 eV and 586.1 eV and are characteristic for chromium $Cr^{(3+)}$ species in Cr_2O_3 oxide. Interestingly, also partly reduced sample (#3) with the lower thickness of the oxide layer still shows the presence of Cr_2O_3 oxide, because no shift in the position of the chromium peaks was observed as compared with the oxidized sample #1. Contrary, after complete oxide reduction (sample #5) a shift in the position of Cr $2p_{3/2}$ and Cr $2p_{1/2}$ peaks appeared to the binding energies of 574.1 eV and 583.4 eV which are characteristic for $Cr^{(0)}$ metal species in the alloy. When comparing spectra for nickel (Fig. 12b) for various treatments, we can see that nickel is always in the same state and it is not affected by plasma treatment.

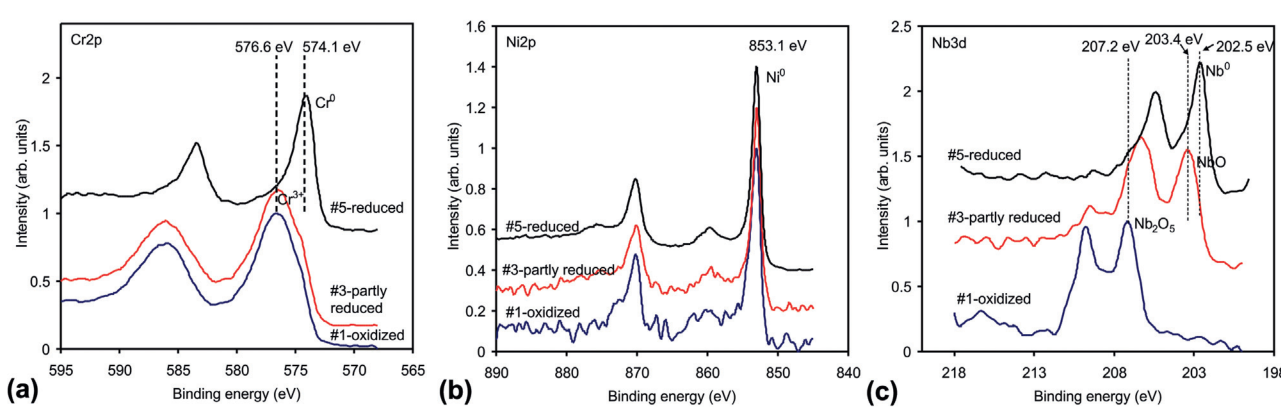

Fig. 12. Comparison of XPS high-resolution spectra of (a) chromium, (b) nickel and (c) niobium for oxidized (#1), partly reduced (#3) and reduced (#5) samples.

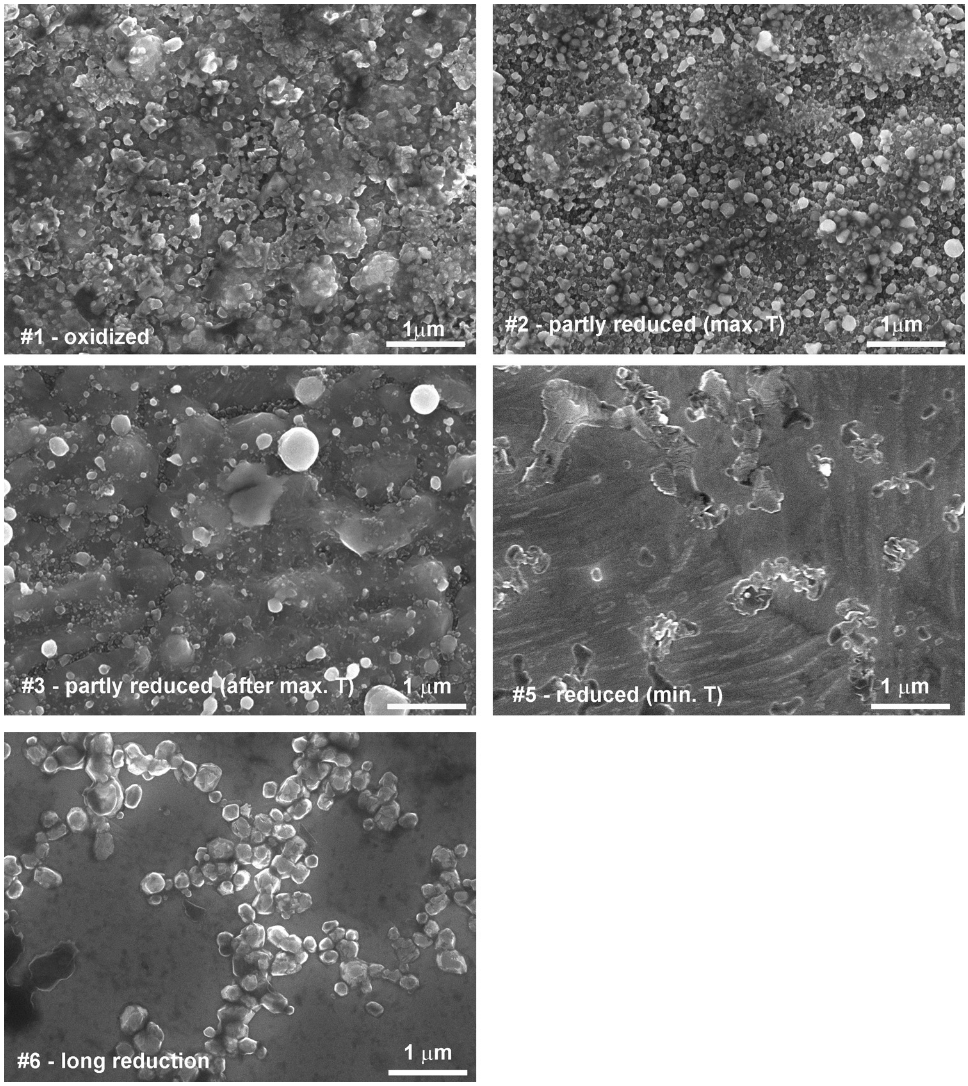

Fig. 14. SEM images of the samples recorded after different reduction steps.

Thus the Ni 2p spectra correspond to the Ni $2p_{3/2}$ and Ni $2p_{1/2}$ peaks located at 853.1 and 870.2 eV which are characteristic of Ni-metal bonds, while the shoulder located at about 860 eV is attributed to the satellite peak of Ni $2p_{3/2}$ feature. Interesting are also results for niobium Nb 3d (Fig. 12c), which is present in the alloy in only small quantities. Here we can see that peaks Nb $3d_{5/2}$ and Nb $3d_{3/2}$ for the oxidized sample (#1) are positioned at 207.2 eV and 209.8 eV and are characteristic for $Nb^{(5+)}$ species in Nb_2O_5 oxide. For partly reduced sample #3, the main peak is shifted to 203.4 eV indicating reduction of $Nb^{(5+)}$ to $Nb^{(2+)}$ and formation of NbO oxide. For reduced sample #5, the peak is further shifted and is positioned at the binding energy characteristic for metallic Nb. Due to completeness of the paper, we also performed XPS depth profiling. Because the sputtering rate in XPS device was just 2 nm/min in comparison to AES depth profiling where it was 10 nm/min, it was not possible to etch through the whole oxide layer but just to a depth of approximately 100 nm. The measured XPS depth profiles are in agreement with the first 10 min of etching in AES depth profiles and are shown in Supplementary information (Fig. S1 in online version at DOI: 10.1016/j.apsusc.2016.06.098).

In order to check any changes in Inconel structure after the reduction experiments, we performed also characterization by X-ray diffraction (XRD). Fig. 13a represents normalized XRD spectra for samples #1–#6 and Fig. 13b a zoom on the two main peaks of Cr_2O_3 (located at 33.6 and 36.2°) for samples #1, #3 and #5. A spectrum for non-oxidized as-received Inconel is also shown (ICDD file 071-7594). The diffraction peaks for all samples are quite similar. The oxidized samples (#1) exhibit oxide peaks of only Cr_2O_3 (ICDD file 73-6214) which are not very intensive due to the small thickness of the oxide film and to the rather large investigation depth of the non-glancing XRD method. The oxide peaks vanish as the reduction progresses, which is sound with the AES depth profiles. As shown in Fig. 13b, a slight decrease in peak intensities can be observed for sample #3 compared to the oxidized sample #1, and no peaks are present for the reduced sample #5. The only changes observed are rather sharp peaks after the reduction in hydrogen plasma at elevated temperature (Fig. 13a). This observation indicates ordering at long range, which is likely due to rather high temperatures involved. This is also in agreement with SEM images (Fig. 14) which show evolution of the surface roughness during the treatment. The oxidized sample is very rough and covered with many small features. During the reduction process, they slowly disappear and only some large crystallites remain. After the oxide removal the surface became smoother with only a few small features, whereas after the longest treatment the remaining small features became more ordered crystallites.

To complete this study let us estimate the real temperatures involved regarding Figs. 4 and 6. Taking into account the emis-

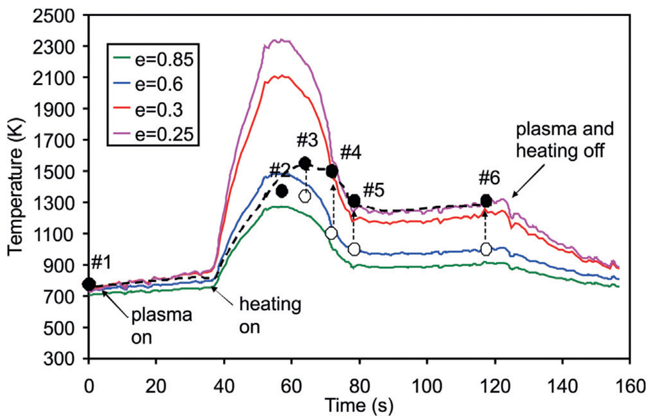

Fig. 15. Most probable temperature of sample upon heating in hydrogen plasma (dashed line).

sivity of well-oxidized Inconel (ε = 0.85), the sample treated in hydrogen without plasma (where no reduction was observed) was around 1500 K. The temperature of the sample treated in plasma was around 1300 K when reduction started. The estimation of final temperature after prolonged treatment (when oxide was completely reduced, sample #6) can be performed taking into account recommended value of emissivity ε = 0.3–0.25 at the pyrometer wavelength i.e. 1210–1270 K. In between, we can speculate that the emissivity changes over a few seconds immediately after reduction of the very surface of our samples so after the first inflection point before the peak in Fig. 6. As already mentioned, we took an intermediate value of the spectral normal emissivity of 0.60 as a rough approximation. The change in emissivity is likely not instantaneous due to the final thickness of reduced layer and perhaps due to lateral heterogeneity. The recommended evolution of the sample temperature versus treatment time is thus presented in Fig. 15 (dashed line). It seems that after reduction starts, the temperature of the sample is still increasing to approximately 1500 K. When reduction terminates, the temperature is settled at approximately 1270 K.

4. Conclusions

Samples of well-oxidized Inconel were efficiently reduced at reasonable low temperature by exposure to low-pressure hydrogen plasma created by a microwave discharge. The reduction kinetic was monitored by measuring the sample radiation using an IR monochromatic pyrometer and by post-mortem characterization using AES depth profiling, XPS and XRD. The reduction does not occur until the temperature of the oxidized sample reaches approximately 1300 K. Once such temperature is exceeded the reduction becomes extensive and the 700 nm thick chromium oxide film is reduced in approximately 10 s. Such a rapid reduction is explained by interaction with neutral hydrogen atoms which can be also stimulated by positively charged hydrogen ions. The composition of the surface film after accomplishing oxide reduction assumes that of bulk Inconel. Such technique therefore represents an ecologically friendly alternative to classical descaling using wet chemical treatments.

Acknowledgements

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